main.c

Run ▶

```c

"include">#include <stdio.h>
"include">#include <conio.h>

"keyword">int "keyword">main()
{
"keyword">int i, NOP, sum = 0, count = 0, y, quant, wt = 0, tat = 0, at[10], bt[10], temp[10];
"keyword">float avg_wt, avg_tat;

printf(" Total number of process in the system: ");
scanf("%d", &NOP);
y = NOP;

"keyword">for(i = 0; i < NOP; i++)
{
    printf("\n Enter the Arrival and Burst time of the Process[%d]\n", i + 1);
    printf(" Arrival time is: \t");
    scanf("%d", &at[i]);
    printf(" \nBurst time is: \t");
    scanf("%d", &bt[i]);
    temp[i] = bt[i];
}

printf("Enter the Time Quantum ">for the process: \t");
scanf("%d", &quant);

printf("\n Process No \t\t Burst Time \t\t TAT \t\t Waiting Time ");

"keyword">for(sum = 0, i = 0; y != 0; )
{
    "keyword">if(temp[i] <= quant && temp[i] > 0)
    {
        sum = sum + temp[i];
        temp[i] = 0;
        count = 1;
    }
    "keyword">else "keyword">if(temp[i] > 0)
    {
        temp[i] = temp[i] - quant;

```

Output

```
 Total number of process in the system: 3

 Enter the Arrival and Burst time of the Process[1]
 Arrival time is:       0
Burst time is:  5

 Enter the Arrival and Burst time of the Process[2]
 Arrival time is:       1
Burst time is:  4

 Enter the Arrival and Burst time of the Process[3]
 Arrival time is:       2
Burst time is:  2
Enter the Time Quantum for the process: 2

 Process No             Burst Time              TAT
Waiting Time
Process No[3]           2
6                        4
Process No[1]           5
11                       6
Process No[2]           4
10                       6
 Average Turn Around Time:      5.333333
 Average Waiting Time:  9.000000

=== Code Execution Successful ===
```

```
    "keyword">if(temp[i] == 0 && count == 1)
    {
        y--;
        printf("\nProcess No[%d] \t\t %d\t\t\t\t %d\t\t\t %d", i + 1, bt[i], sum - at[i], sum - at[i] - bt[i]);
        wt = wt + sum - at[i] - bt[i];
        tat = tat + sum - at[i];
        count = 0;
    }

    "keyword">if(i == NOP - 1)
    {
        i = 0;
    }
    "keyword">else "keyword">if(at[i + 1] <= sum)
    {
        i++;
    }
    "keyword">else
    {
        i = 0;
    }
}

avg_wt = wt * 1.0 / NOP;
avg_tat = tat * 1.0 / NOP;

printf("\n Average Turn Around Time: \t%f", avg_wt);
printf("\n Average Waiting Time: \t%f", avg_tat);

getch();
}
```